BABU BANARASI DAS UNIVERSITY

LUCKNOW

BCA (CYBER SECURITY & FORENSICS)

4TH SEMSTER

BBD UNIVERSITY

SUBJECT - APPLY END TO END SECURITY TO CLOUD APPLICATION

PROJECT

SUBMITTED TO

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Project :-

A healthcare company stores sensitive patient data in the cloud and analyze how data anonymization, tokenization and data loss prevention can be used to protect data, identify possible risks, and solution.

1. Data Anonymization :-
Removing or modifying personal identifier so that individuals cannot be identified.

Ex:- Replacing patient names, phone number, or Aadhar numbers with random values or removing them entirely.

How it protect data :
- Even if data is leaked, it cannot be traced back to a specific person.
- useful for research and analytics without exposing identity.

Risk :-
- Poor anonymization techniques can leave patterns
- Re-identification risk: Attackers may combine anonymized data with other datasets to identify individual.

Solution :-
- use strong anonymization methods like k-anonymity or differential privacy.
- Regular test dataset for reidentification risks.
- Limit access to raw (original) data.

Tokenization :-

Replacing sensitive data with a unique token (random string) while storing the original data securely elsewhere.

Example :- Patient ID → replaced with token like 'X9A7K2'

How it protects data :-

- Even if attackers get tokens, they are meaningless without access to the secure mapping system.
- Reduces exposure of real data in applications & databases.

Risk :-

- If the token vault (where original data is stored) is compromised, all data can be exposed.
- Poor key management can weaken security.

Solution :-

- Use strong encryption for the token vault.
- Apply strict access control & authentication.
- Regularly rotate encryption keys & monitor access logs.

Data loss Prevention (DLP) :-

Security tools and policies that direct & prevent unauthorized sharing or leakage of sensitive data.

Example :- Blocking an employee from emailing patient records outside the organization.

Method :-

- Monitoring data access & usage.
- Content inspection & filtering
- Incident response & remediation

Benefits :- Detects & prevents data breaches, ensures & compliances

Risks :- Complex setup, potential for false positives.

Section :- 2

A cloud based organization frequent network attack design a secure architecture using SDN, VLANs & IDS.

Secure Architecture design :-

Network Segmentation :-

VLAN.	Purpose	Security Measures
VLAN10	web	Public facing, restricted access.
VLAN20	App	Application Server, limited access
VLAN 30	DB	Sensitive data, restricted access.

SDN configuration :-

SDN controller :- Manages network policies automate security rule

Flow rules :-

Allow web traffic VLAN 10.

Restrict DB traffic from VLAN 20 to VLAN30

show DB traffic from VLAN 20 to VLAN 30

IDS configuration :-

• I.D.S placement :- deploy IDS sensor in each VLAN

Rules :-

Monitor for suspicious traffic patterns

Alert on potential threats.